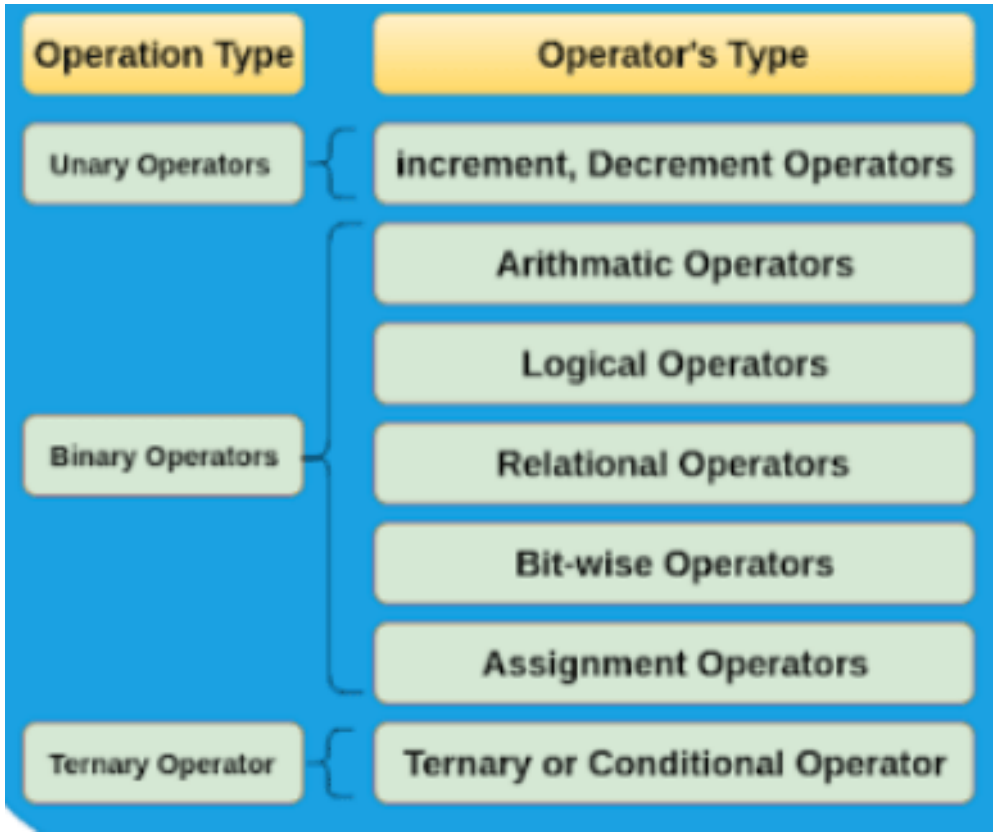
BCSE102L Structured and Object Oriented Programming

Operators in C

C supports a rich set of built-in operators.

An operator is a symbol that tells the computer to perform certain mathematical or logical manipulations.



Operators in C

Arithmetic Operators

Operates on any built-in data types

Example operators:

1) **+ (Plus) –** Give sum as a result. Example: 3+2=5
2) **– (Minus)** – Give the difference as a result. Example: 4-2=2
3) *** (Asterisk)** – Used for multiplication and give the product as a result. Example: 4*2=8
4) **/ (Forward slash)** – Used for division and give quotient as a result. Example: 4/2=2
5) **% (modulus)** – Give the remainder as a result. Example: 7%2=1

Operators in C

Relational Operators

To compare two quantities, and depending on their relation, take certain decisions.

Eg. a>b

Operator	Use
<	Less than
<=	Less than or equal to
>	Greater than
>=	Greater than or equal to
==	Equal
!=	Not equal

Operators in C

Logical Operators

The logical operators are used to test more than one conditions and make decisions.

Example Operators:

Logical AND	&&
Logical OR	||
Logical NOT	!

Eg. a>b && a>c

Operators in C

Assignment Operators

It is used to assign the results of an expression to a variable.

Syntax:

var op = exp;

var = var op (exp);

Example:

x += y+1

x = x + (y+1);

var – variable, op – binary arithmetic operator, exp - expression

Operators in C

Increment and Decrement Operators

1. Increment operators (++): Example: (++x)
2. Decrement operators (–): Example: (--m)

These operators can be prefix operator or postfix operator.

Prefix increment operator: ++m is equivalent to m=m+1 or m+=1

Postfix increment operator: a = m++ -j+10, in this case the expression is evaluated first using the original value of the variable and then variable is incremented.

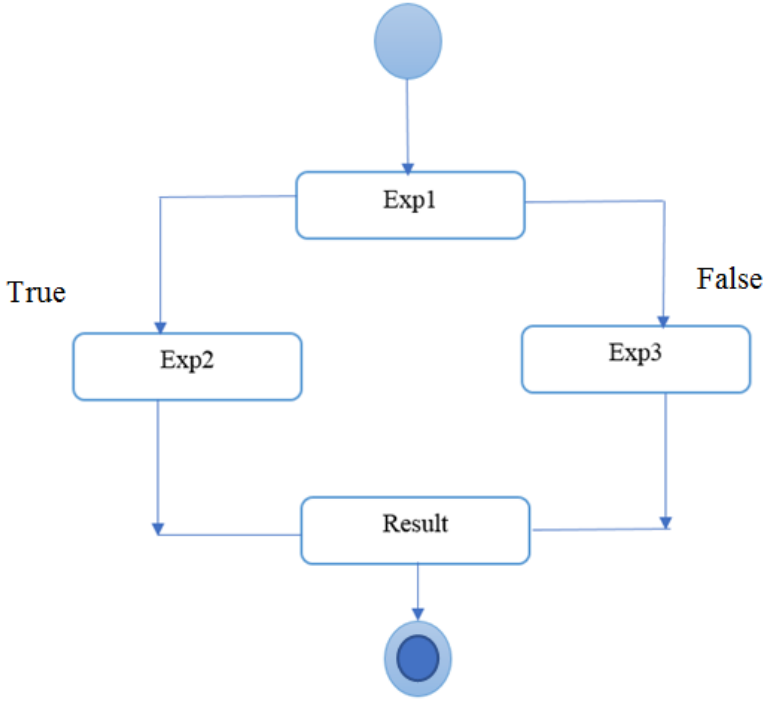
Operators in C

Conditional Operator

A ternary operator pair ":?" is available in C to construct conditional expressions of the form:

Syntax: exp1 ? exp2 : exp3

Example: x = (a>b) ? a : b;



Operators in C

Bitwise Operator

Used for manipulation of data's at bit level.

Used for testing the bits, or shifting them right or left

operator's	meaning of operators
&	Bitwise AND
|	Bitwise OR
^	Bitwise XOR
~	Bitwise Complement
<<	Bitwise left shift
>>	Bitwise right shift

Operators in C

Special Operators

1. The comma operator – used to link the related expressions together
   Example: value = (x=10, y=5, x+5);
2. The sizeof operator – returns the number of bytes the operand occupies
   Example: m = sizeof(sum);
3. Pointer operator (& and *)
4. Member selection operator (. and ->)

Operator Precedence and Associativity

C has precedence associated with it.
This precedence is used to determine how an expression involving more than one operator is evaluated

The operators at the highest level are evaluated first
The operators at the same precedence are evaluated either from left to right or from right to left, depending on the level. This is known as associativity property of an operator.

Consider the conditional statement,

if (x == 10 + 15 && y < 10)

if (x == 25 && y < 10)

// Precedence rule: addition operator has highest priority than logical and relational operators

Precedence of Arithmetic Operators

Category	Operator	Associativity
Postfix	() [] -> . ++ - -	Left to right
Unary	+ - ! ~ ++ - - (type)* &sizeof	Right to left
Multiplicative	* / %	Left to right
Additive	+ -	Left to right
Shift	<<>>	Left to right
Relational	<<= >>=	Left to right
Equality	== !=	Left to right
Bitwise AND	&	Left to right
Bitwise XOR	^	Left to right
Bitwise OR	|	Left to right
Logical AND	&&	Left to right
Logical OR	||	Left to right
Conditional	?:	Right to left
Assignment	= += -= *= /= %=>>= <<= &= ^= |=	Right to left
Comma	,	Left to right

Precedence of Arithmetic Operators

An example:

```
#include <stdio.h>
int main()
{
float a,b,c,x;
a=9; b=12; c=3;
x= a - b / 3 + c * 2 - 1;
printf("x=%f", x);
return 0;
}
```

OUTPUT?

How to read data from keyboard?

Scanf(“control string”, &variable1, &variable2,..);

Scanf(“%d”, &number);

Example:

```
#include <stdio.h>
int main()
{
int a,b,sum;
printf("Enter the value for A and B: ");
scanf("%d %d", &a, &b);
sum=a+b;
printf("Sum=%d", sum);
return 0;
}
```

Simple C Programs

Program: To print a text

```
#include <stdio.h>
int main()
{
printf("Hello, World! \n");
return 0;
}
```

Program: Display Variable Value

```
#include <stdio.h>
int main()
{
    int a=50;
    int b=5;
    //a=50, b=5;
    int c=a+b;
    printf("Sum of %d and %d is: %d", a, b, c);
}
```

```
#include <stdio.h>
main()
{
int number;
float amount;
number=100;
amount=32.75+53.50;
printf("%d", number);
printf("\n%4.2f", amount);
  //return 0;
}
```

Program: Find the area of the square

```
#include<stdio.h>
 int main() {
   int side, area;
    printf("\n Enter the Length of Side : ");
   scanf("%d", &side);
    area = side * side;
   printf("\n Area of Square : %d", area);
    return (0);
}
```

Program to calculate simple interest.

```
#include<stdio.h>
#include<conio.h>
int main()
{
float p,rate,time,si;
 printf("Enter principal amount : ");
 scanf("%f", &p);
 printf("Enter rate of interest : ");
 scanf("%f", &rate);
 printf("Enter time period in  year : ");
 scanf("%f", &time);
/*calculating simple interest*/
 si=(p*time*rate)/100;
 printf("\nSimple Interest = %f",si);
    getch();
 return 0;}
```

Program to swap two numbers.

```
#include <stdio.h>
int main()
{
   int A, B, temp;
   printf("Please enter the 1st number : ");
   scanf("%d",&A);
   printf("\nPlease enter the 2nd number : ");
   scanf("%d",&B);
   printf("\nBefore swapping:\n");
   printf("A - %d \nB - %d", A, B);

  temp = A;
   A = B;
   B = temp;

printf("\nAfter swapping:\n");
   printf("A - %d \nB - %d", A, B);
   return 0;
}
```